e) Implementation of K-Means Algorithm

Aim:

To use K-Means to cluster data points into groups (e.g., customers or flowers).

Procedure:

1. Load data.
2. Fit KMeans.
3. Plot or evaluate clusters.

Python Code:

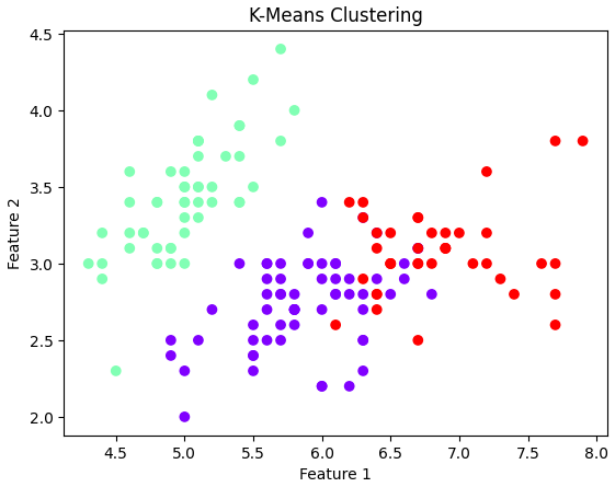
```
from sklearn.datasets import load_iris
from sklearn.cluster import KMeans
import matplotlib.pyplot as plt

# Load data
data = load_iris()
X = data.data

# Apply KMeans
kmeans = KMeans(n_clusters=3, random_state=0)
kmeans.fit(X)
labels = kmeans.labels_

# Plot (2 features only for visualization)
plt.scatter(X[:, 0], X[:, 1], c=labels, cmap='rainbow')
plt.xlabel("Feature 1")
plt.ylabel("Feature 2")
plt.title("K-Means Clustering")
plt.show()
```

Output:



Result:

Thus the K-Means clustering algorithm was implemented and the clusters were visualized successfully.